US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12412649
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Steinhauer; Tom et al.

Ventilation management system

Abstract

A ventilation management system communicatively couples a remote device such as a mobile device to one or more ventilators for monitoring of ventilator data, including ventilator configuration data, patient physiological statistics, and notifications. When a command to modify a configuration parameter of the ventilator is received while the ventilator is unavailable for communication, the command is cached and provided when the ventilator becomes available. When ventilator becomes available, the system receives an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable. The operating condition of the ventilator, the physiological statistic of the patient, and the indication that the configuration parameter was modified is provided to the remote computing device for display.

Inventors: **Steinhauer; Tom (San Diego, CA), Lam; Willis (San Diego, CA), Rogers; Mark (Irvine, CA), Blansfield; Terry (Orange, CA), Birch; Stephen J (Mission Viejo, CA)**

Applicant: **ZOLL MEDICAL CORPORATION** (Chelmsford, MA)

Family ID: **48901799**

Assignee: **ZOLL MEDICAL CORPORATION (Chelmsford, MA)**

Appl. No.: **18/297572**

Filed: **April 07, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230245755 A1	Aug. 03, 2023

Related U.S. Application Data

continuation parent-doc US 16857060 20200423 US 11626199 child-doc US 18297572
continuation parent-doc US 15394346 20161229 US 10646674 20200512 child-doc US 16857060
continuation parent-doc US 13830730 20130314 US 9821129 20171121 child-doc US 15394346
continuation-in-part parent-doc US 13287419 20111102 ABANDONED child-doc US 13830730

Publication Classification

Int. Cl.: **G16H20/40** (20180101); **A61M16/00** (20060101); **A61M16/10** (20060101); **G16H10/60** (20180101); **G16H40/63** (20180101); **G16H40/67** (20180101)

U.S. Cl.:

CPC **G16H20/40** (20180101); **A61M16/0051** (20130101); **A61M16/0057** (20130101); **A61M16/021** (20170801); **A61M16/026** (20170801); **A61M16/1005** (20140204); **G16H10/60** (20180101); **G16H40/63** (20180101); **G16H40/67** (20180101); A61M2205/3553 (20130101); A61M2205/3561 (20130101); A61M2205/3584 (20130101); A61M2205/3592 (20130101); A61M2205/502 (20130101); A61M2205/505 (20130101); A61M2205/52 (20130101); A61M2205/6009 (20130101); A61M2205/6018 (20130101); A61M2205/6054 (20130101); A61M2205/6072 (20130101); A61M2205/609 (20130101); A61M2230/005 (20130101); A61M2230/202 (20130101); A61M2230/42 (20130101); A61M2230/432 (20130101); A61M2230/435 (20130101); A61M2230/46 (20130101)

Field of Classification Search

CPC: A61M (16/00); A61M (16/0057); A61M (2205/6009); A61M (2205/3584); A61M (2205/505); A61M (2205/609); A61M (2205/3553); A61M (2205/6087); A61M (2205/6072); A61M (2205/6018); A61M (2230/435); A61M (2230/432); A61M (2230/202)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4988336	12/1990	Kohn	N/A	N/A
5065315	12/1990	Garcia	N/A	N/A
5544649	12/1995	David et al.	N/A	N/A
5626151	12/1996	Linden	N/A	N/A
5881723	12/1998	Wallace et al.	N/A	N/A
5931160	12/1998	Gilmore et al.	N/A	N/A
5987519	12/1998	Peifer et al.	N/A	N/A
6148814	12/1999	Clemmer et al.	N/A	N/A
6158430	12/1999	Pfeiffer	128/204.21	A61M 16/00
6158433	12/1999	Ong et al.	N/A	N/A
6349724	12/2001	Burton	N/A	N/A

6369847	12/2001	James et al.	N/A	N/A
6406426	12/2001	Reuss et al.	N/A	N/A
6459933	12/2001	Lurie et al.	N/A	N/A
6551243	12/2002	Bocionek	128/923	G16H 70/40
6666820	12/2002	Poole	N/A	N/A
6839753	12/2004	Biondi et al.	N/A	N/A
6955170	12/2004	Mullins et al.	N/A	N/A
7165221	12/2006	Monteleone et al.	N/A	N/A
7225809	12/2006	Bowen et al.	N/A	N/A
7237205	12/2006	Sarel	N/A	N/A
7334578	12/2007	Biondi et al.	N/A	N/A
7395216	12/2007	Rosenfeld et al.	N/A	N/A
7610099	12/2008	Almendinger et al.	N/A	N/A
7668596	12/2009	Von Arx et al.	N/A	N/A
7677246	12/2009	Kepler et al.	N/A	N/A
7698156	12/2009	Martucci et al.	N/A	N/A
7933780	12/2010	De La Huerga	N/A	N/A
8015972	12/2010	Pirzada	N/A	N/A
8255238	12/2011	Powell et al.	N/A	N/A
8321284	12/2011	Clements et al.	N/A	N/A
8327846	12/2011	Bowditch et al.	N/A	N/A
8447629	12/2012	Rappaport et al.	N/A	N/A
8522779	12/2012	Lee et al.	N/A	N/A
8695593	12/2013	Tehrani	N/A	N/A
9058741	12/2014	Steinhauer et al.	N/A	N/A
9072849	12/2014	Steinhauer et al.	N/A	N/A
9177109	12/2014	Steinhauer et al.	N/A	N/A
9327090	12/2015	Steinhauer	N/A	N/A
9352110	12/2015	Steinhauer	N/A	N/A
9687618	12/2016	Steinhauer et al.	N/A	N/A
9737676	12/2016	Steinhauer et al.	N/A	N/A
9821129	12/2016	Steinhauer et al.	N/A	N/A
10646674	12/2019	Steinhauer et al.	N/A	N/A
2001/0016821	12/2000	DeBusk et al.	N/A	N/A
2001/0027791	12/2000	Wallace et al.	N/A	N/A
2002/0022973	12/2001	Sun et al.	N/A	N/A
2002/0026941	12/2001	Biondi et al.	N/A	N/A
2002/0077862	12/2001	Auer et al.	N/A	N/A
2002/0091309	12/2001	Auer	N/A	N/A
2002/0120676	12/2001	Biondi et al.	N/A	N/A
2002/0133061	12/2001	Manetta	N/A	N/A
2003/0050802	12/2002	Jay et al.	N/A	N/A
2003/0101076	12/2002	Zaleski	N/A	N/A
2004/0077934	12/2003	Massad	N/A	N/A
2004/0249675	12/2003	Stark et al.	N/A	N/A
2005/0108057	12/2004	Cohen et al.	N/A	N/A
2005/0137653	12/2004	Friedman et al.	N/A	N/A
2005/0143632	12/2004	Elaz	N/A	N/A
2005/0151640	12/2004	Hastings	N/A	N/A
2005/0188083	12/2004	Biondi et al.	N/A	N/A

2005/0192845	12/2004	Brinsfield et al.	N/A	N/A
2005/0267348	12/2004	Wollenweber et al.	N/A	N/A
2005/0268916	12/2004	Mumford et al.	N/A	N/A
2006/0031095	12/2005	Barth et al.	N/A	N/A
2006/0144401	12/2005	Boelt	N/A	N/A
2006/0162727	12/2005	Biondi et al.	N/A	N/A
2006/0174883	12/2005	Aylsworth et al.	N/A	N/A
2006/0180150	12/2005	Dittmann	N/A	N/A
2006/0189854	12/2005	Webb et al.	N/A	N/A
2006/0206011	12/2005	Higgins et al.	N/A	N/A
2006/0289020	12/2005	Tabak et al.	N/A	N/A
2007/0005621	12/2006	Lesh et al.	N/A	N/A
2007/0015976	12/2006	Miesel et al.	N/A	N/A
2007/0023045	12/2006	Kwok et al.	N/A	N/A
2007/0033072	12/2006	Bildirici	N/A	N/A
2007/0227537	12/2006	Bemister	700/84	A61M 16/021
2007/0276696	12/2006	Gauvin et al.	N/A	N/A
2007/0283958	12/2006	Miesel et al.	N/A	N/A
2008/0000479	12/2007	Elaz	128/204.23	A61M 16/0003
2008/0004904	12/2007	Tran	340/286.07	G16H 40/67
2008/0053438	12/2007	DeVries et al.	N/A	N/A
2008/0072902	12/2007	Setzer et al.	N/A	N/A
2008/0077436	12/2007	Muradia	N/A	N/A
2008/0086691	12/2007	Hopermann et al.	N/A	N/A
2008/0091466	12/2007	Butler et al.	N/A	N/A
2008/0097793	12/2007	Dicks et al.	N/A	N/A
2008/0097913	12/2007	Dicks et al.	N/A	N/A
2008/0140160	12/2007	Goetz et al.	N/A	N/A
2008/0230064	12/2007	Tham	N/A	N/A
2008/0236582	12/2007	Tehrani	128/204.22	A61M 16/024
2008/0271736	12/2007	Leonard et al.	N/A	N/A
2008/0288023	12/2007	John	N/A	N/A
2008/0308101	12/2007	Spandorfer	N/A	N/A
2008/0312548	12/2007	Hartley et al.	N/A	N/A
2009/0044803	12/2008	Garcia Fernandez	N/A	N/A
2009/0054735	12/2008	Higgins	N/A	N/A
2009/0112160	12/2008	Yang	N/A	N/A
2009/0171696	12/2008	Allard et al.	N/A	N/A
2009/0184823	12/2008	Tessier	N/A	N/A
2009/0229610	12/2008	Oates et al.	N/A	N/A
2009/0241956	12/2008	Baker, Jr.	128/204.23	A61M 16/12
2009/0293886	12/2008	Dedrick et al.	N/A	N/A
2009/0326389	12/2008	Ralfs	N/A	N/A
2010/0071697	12/2009	Jafari et al.	N/A	N/A
2010/0078017	12/2009	Andrieux et al.	N/A	N/A

2010/0083968	12/2009	Wondka et al.	N/A	N/A
2010/0085156	12/2009	Tucker	N/A	N/A
2010/0108064	12/2009	Blackwell et al.	N/A	N/A
2010/0161345	12/2009	Cain et al.	N/A	N/A
2010/0202354	12/2009	Ho	N/A	N/A
2010/0287006	12/2009	Cannon	455/414.1	G06Q 10/06
2010/0288279	12/2009	Seiver et al.	N/A	N/A
2010/0298718	12/2009	Gilham et al.	N/A	N/A
2010/0318155	12/2009	Mahajan et al.	N/A	N/A
2010/0326446	12/2009	Behlmaier	N/A	N/A
2011/0011400	12/2010	Gentner et al.	N/A	N/A
2011/0034783	12/2010	Lisogurski	600/301	A61B 5/0002
2011/0073107	12/2010	Rodman	141/2	G16H 40/63
2011/0077970	12/2010	Mellin et al.	N/A	N/A
2011/0078253	12/2010	Chan et al.	N/A	N/A
2011/0087756	12/2010	Biondi et al.	N/A	N/A
2011/0107251	12/2010	Guaitoli et al.	N/A	N/A
2011/0108034	12/2010	Viertio-Oja	N/A	N/A
2011/0112442	12/2010	Meger et al.	N/A	N/A
2011/0120470	12/2010	Bowerbank	128/204.23	G06Q 10/06
2011/0139155	12/2010	Farrell et al.	N/A	N/A
2011/0178373	12/2010	Pacey et al.	N/A	N/A
2011/0208539	12/2010	Lynn	N/A	N/A
2011/0231505	12/2010	Chan et al.	N/A	N/A
2011/0238441	12/2010	Callas	N/A	N/A
2011/0319322	12/2010	Bashan et al.	N/A	N/A
2012/0082036	12/2011	Abedi et al.	N/A	N/A
2012/0108984	12/2011	Bennett et al.	N/A	N/A
2012/0109240	12/2011	Zhou et al.	N/A	N/A
2012/0118285	12/2011	Wondka et al.	N/A	N/A
2012/0137250	12/2011	Milne et al.	N/A	N/A
2012/0145153	12/2011	Bassin et al.	N/A	N/A
2012/0216809	12/2011	Milne et al.	N/A	N/A
2012/0272955	12/2011	Cool et al.	N/A	N/A
2012/0330177	12/2011	Al-Rawas et al.	N/A	N/A
2013/0032147	12/2012	Robinson et al.	N/A	N/A
2013/0054264	12/2012	Baronov et al.	N/A	N/A
2013/0104889	12/2012	Steinhauer et al.	N/A	N/A
2013/0104890	12/2012	Steinhauer et al.	N/A	N/A
2013/0104891	12/2012	Steinhauer et al.	N/A	N/A
2013/0104892	12/2012	Steinhauer et al.	N/A	N/A
2013/0104893	12/2012	Steinhauer et al.	N/A	N/A
2013/0104894	12/2012	Steinhauer et al.	N/A	N/A
2013/0104895	12/2012	Steinhauer et al.	N/A	N/A
2013/0104897	12/2012	Steinhauer et al.	N/A	N/A
2013/0106864	12/2012	Boyer	N/A	N/A
2013/0110528	12/2012	Steinhauer et al.	N/A	N/A
2013/0110529	12/2012	Steinhauer et al.	N/A	N/A
2013/0110530	12/2012	Steinhauer et al.	N/A	N/A

2013/0110546	12/2012	Steinhauer et al.	N/A	N/A
2013/0110924	12/2012	Steinhauer et al.	N/A	N/A
2013/0186405	12/2012	Krzyzanowski et al.	N/A	N/A
2013/0199533	12/2012	Steinhauer et al.	N/A	N/A
2014/0000604	12/2013	Steinhauer et al.	N/A	N/A
2014/0000605	12/2013	Steinhauer et al.	N/A	N/A
2014/0000607	12/2013	Steinhauer et al.	N/A	N/A
2014/0000608	12/2013	Steinhauer et al.	N/A	N/A
2014/0002246	12/2013	Steinhauer et al.	N/A	N/A
2014/0006041	12/2013	Steinhauer et al.	N/A	N/A
2014/0006052	12/2013	Steinhauer et al.	N/A	N/A
2014/0158124	12/2013	L'her et al.	N/A	N/A
2015/0182712	12/2014	Kelly et al.	N/A	N/A
2019/0184119	12/2018	Crawford et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109661197	12/2018	CN	N/A
2181726	12/2009	EP	N/A
2773410	12/2013	EP	N/A
3503795	12/2018	EP	N/A
2019528153	12/2018	JP	N/A
WO0244993	12/2001	WO	N/A
WO2011087111	12/2010	WO	N/A
WO-2013067223	12/2012	WO	N/A
WO-2018038853	12/2017	WO	N/A

OTHER PUBLICATIONS

“Medicare Quarterly Provider Compliance Newsletter”, Oct. 2011, 21 pages, vol. 2, Issue 1, CMS. cited by applicant
“Solutions,” retrieved on Sep. 16, 2013, 1 page, Theronyx, retrieved from <http://www.theronyx.com/solutions>. cited by applicant
“Vital Sync Virtual Patient monitoring Platform 2.0,” retrieved on Sep. 16, 2013, 2 pages, Covidien, retrieved from <http://www.covidien.com/RMS/pages.aspx?page+Product/Vital-Sync-Vitrual-Patient-Monitoring-Platform>. cited by applicant
Canadian Office Action for Application No. 2904004, dated Aug. 10, 2020, 4 pages. cited by applicant
Canadian Office Action for Application No. 2904004, dated Jul. 3, 2019, 5 pages. cited by applicant
Chipman, Daniel W., et al. “Performance comparison of 15 transport ventilators.” Respiratory care 52.6 (2007): 740-751. cited by applicant
European Office Action for Application No. 12844716.6, dated Nov. 6, 2017, 7 pages. cited by applicant
European Office Action for Application No. 20151238.1, dated Mar. 21, 2022, 4 pages. cited by applicant
Extended European Search Report for Application No. 12844716.6, dated Jan. 4, 2016, 12 pages. cited by applicant
Extended European Search Report for Application No. 13878057.2, dated Oct. 4, 2016, 8 pages.

cited by applicant
Extended European Search Report for Application No. 20151238.1, dated May 15, 2020, 7 pages. cited by applicant
Extended European Search Report for Application No. 21187978.9, dated Nov. 24, 2021, 11 pages. cited by applicant
Fairley, H. Barrie, and B.A. Britt. “The adequacy of the air-mix control in ventilators operated from an oxygen source.” Canadian Medical Association Journal 90.25 (1964): 1394. cited by applicant
Govoni, et al., “An Improved Telemedicine System for Remote Titration and Optimization of Home Mechanical Ventilation”, Biomedical Engineering Conference (CIBEC), 2010 5th Cairo International, IEEE, Dec. 16, 2010, pp. 66-69, XP031979754. cited by applicant
International Preliminary Report of Patentability for International Application No. PCT/US2013/048734, dated Dec. 31, 2014, 8 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/US2012/063119, dated Feb. 27, 2014, 14 pages. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2012/063119, dated Mar. 29, 2013, 8 pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2013/048734, dated Sep. 9, 2013, 12 pages. cited by applicant
International Search Report for International Application No. PCT/US2013/057860, dated Nov. 7, 2013, 2 pages. cited by applicant
International Search Report for International Application No. PCT/US2013/057862, dated Dec. 12, 2013, 2 pages. cited by applicant
Krieger, Bruce P., et al. “Initial experience with a central respiratory monitoring unit as a cost-saving alternative to the intensive care unit for Medicare patients who require long-term ventilator support.” Chest Journal 93.2 (1988): 395-397. cited by applicant
Shimpi, A.L., “iphone 3GS Performance: A Significant Performance Bump,” Anand Tech (2009), http://www.anandtech.com/show/2790. cited by applicant
Simonds, A. K. “Streamlining weaning: protocols and weaning units.” Thorax 60.3 (2005): 175-182. cited by applicant
Title: CPAP Equipment: CPAP Software Date Archived: May 12, 2012 Publisher: cpap.com. cited by applicant
Extended European Search Report for Application No. 25156085.0, dated May 23, 2025, 9 pages. cited by applicant

---

Primary Examiner: Stuart; Colin W

Assistant Examiner: Sul; Douglas Y

Attorney, Agent or Firm: MASCHOFF BRENNAN

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of U.S. patent application Ser. No. 16/857,060, entitled ‘VENTILATION MANAGEMENT SYSTEM”, filed on Apr. 23, 2020, now U.S. Pat. No. 11,626,199, which is a continuation application of U.S. patent application Ser. No. 15/394,346, entitled “VENTILATION MANAGEMENT SYSTEM,” filed on Dec. 29, 2016, now U.S. Pat. No. 10,646,674, which is a continuation application of U.S. patent application Ser. No. 13/830,730, entitled “VENTILATION MANAGEMENT SYSTEM,” filed on Mar. 14, 2013, now U.S. Pat. No. 9,821,129, which is a continuation-in-part application of U.S. patent application Ser. No. 13/287,419, entitled “BI-DIRECTIONAL VENTILATOR COMMUNICATION,” filed on Nov. 2, 2011. The disclosures of these applications is hereby incorporated by reference in their entirety for all purposes.

BACKGROUND

Field

(1) The present disclosure generally relates to medical devices, and more particularly to the configuration of a ventilator.

Description of the Related Art

(2) Medical ventilation systems (or “ventilators,” colloquially called “respirators”) are machines that are typically used to mechanically provide breathable air or blended gas to lungs in order to assist a patient in breathing. Ventilation systems are chiefly used in intensive care medicine, home care, emergency medicine, and anesthesia. Common ventilation systems are limited to a single direction of communication, and as such are configured to provide information related to the ventilation system for display, but not receive information from a remote source for any purpose to control the ventilator. For example, common ventilation systems send outbound data to another entity, such as a display device, in order to display ventilator settings.

SUMMARY

(3) According to certain embodiments of the present disclosure, a ventilation system is provided. The system includes a ventilation device that is configured to provide breathable air to a patient according to certain operating parameters, a memory that includes instructions, and a processor. The processor is configured to execute the instructions to receive, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and provide a modification of operating parameters of the ventilation device based on the received patient data, order data, configuration data, user data, or rules or protocols.

(4) According to certain embodiments of the present disclosure, a method for configuring a ventilator is provided. The method includes receiving, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and providing a modification of operating parameters of a ventilation device that is configured to provide breathable air to a patient according to the operating parameters based on the received patient data, order data, configuration data, user data, or rules or protocols.

(5) According to certain embodiments of the present disclosure, a machine-readable storage medium includes machine-readable instructions for causing a processor to execute a method for configuring a ventilator is provided. The method includes receiving, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and providing a modification of operating parameters of a ventilation device that is configured to provide breathable air to a patient according to the operating parameters based on the received patient data, order data, configuration data, user data, or rules or protocols.

(6) According to certain embodiments of the present disclosure, a ventilator management system is provided. The system includes a memory that includes an initial configuration profile configured to designate operating parameters for a ventilation device, and a processor. The processor is configured to receive ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or physiological statistics of a patient associated with the ventilation device, and determine, based on the ventilator data, a modification to the initial configuration profile for the ventilation device. The processor is also configured to generate a modified configuration profile for the ventilation device based on the determined modification.

(7) According to certain embodiments of the present disclosure, a method for managing a plurality of ventilators is provided. The method includes receiving ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or

physiological statistics of a patient associated with the ventilation device, and determining, based on the ventilator data, a modification to an initial configuration profile for the ventilation device. The method also includes generating a modified configuration profile for the ventilation device based on the determined modification.
(8) According to certain embodiments of the present disclosure, a machine-readable storage medium includes machine-readable instructions for causing a processor to execute a method for managing a plurality of ventilators is provided. The method includes receiving ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or physiological statistics of a patient associated with the ventilation device, and determining, based on the ventilator data, a modification to an initial configuration profile for the ventilation device. The method also includes generating a modified configuration profile for the ventilation device based on the determined modification.
(9) It is understood that other configurations of the subject technology will become readily apparent to those skilled in the art from the following detailed description, wherein various configurations of the subject technology are shown and described by way of illustration. As will be realized, the subject technology is capable of other and different configurations and its several details are capable of modification in various other respects, all without departing from the scope of the subject technology. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are included to provide further understanding and are incorporated in and constitute a part of this specification, illustrate disclosed embodiments and together with the description serve to explain the principles of the disclosed embodiments. In the drawings:
(2) FIG. **1** illustrates an example architecture for a ventilator management system.
(3) FIG. **2** is a block diagram illustrating an example ventilation system, ventilation management system, and home ventilation device from the architecture of FIG. **1** according to certain aspects of the disclosure.
(4) FIG. **3** illustrates an example flow chart of exchanging data between a ventilation system and a ventilation management system.
(5) FIG. **4** illustrates an example flow chart for a communication protocol used by the ventilation system of FIG. **3**.
(6) FIG. **5** illustrates example processes for contextualizing ventilator data for a ventilation system.
(7) FIGS. **6**A and **6**B illustrate example flow charts for caching data on a ventilation system and a ventilation management system.
(8) FIG. **7** illustrates an example process for managing a ventilation system.
(9) FIG. **8** is a block diagram illustrating an example computer system with which the example ventilation system, ventilation management system, and home ventilation device of FIG. **2** can be implemented.
DETAILED DESCRIPTION
(10) In the following detailed description, numerous specific details are set forth to provide a full understanding of the present disclosure. It will be apparent, however, to one ordinarily skilled in the art that the embodiments of the present disclosure may be practiced without some of these specific details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the disclosure.
(11) Certain aspects of the disclosed system provide ventilation systems with two-way

communication. Specifically, in addition to permitting a ventilation system to output basic ventilation data such as physiological statistics, the disclosed ventilation systems permit output of additional information such as ventilator settings, notifications, patient information, ventilation waveforms, loops or trended data ("scalars"), and ventilation monitoring information. The disclosed ventilation systems also permit input of configuration profiles, rules and clinical protocols, user data, notifications, preprogramming, patient data, and lab results. The disclosed ventilation systems are configured to operate according to the received configuration profiles, rules, and protocols, and in view of the user data, notifications, preprogramming, patient data, and lab results. The data for the ventilation system can also be "contextualized" (e.g., associated with a patient and/or caregiver) using various wired and wireless techniques. The disclosed ventilation systems are configured to provide the output of additional information to, for example, a ventilation management system.

(12) The disclosed ventilation management system is configured to receive the information from one or many ventilation systems, analyze the information, and determine new or modified configuration profiles, rules, and clinical protocols from the received information. The information may be received wired or wirelessly over a network. The disclosed ventilation management system is also configured to provide the new or modified configuration profiles, rules, and clinical protocols back to one or many of the ventilation systems. The ventilation systems managed by the ventilation management system can be located either in a healthcare institution (e.g., a hospital) or outside of a healthcare institution (e.g., a home or other care site). Both the ventilation systems and the ventilation management systems are configured to cache data, for example, when the network is not available, so that data may be saved for later transmission.

(13) FIG. **1** illustrates an example architecture **10** for a ventilator management system. The architecture **10** includes a ventilation system **102** and a hospital ventilation management system **14** connected over a local area network (LAN) **119** in a hospital **101**, and a home ventilation device **130** in a home **140** connected to a wide area ventilation management system **16** over a wide area network (WAN) **120**. The hospital ventilation management system **14**, which can be configured, for example, by a clinician **12**, other healthcare provider, or administrator, is connected to the wide area ventilation management system **16** through the WAN **120**. Furthermore, the home ventilation device **130** may operate substantially similar to, and be configured substantially the same as, the ventilation system **102** of the hospital **101**, except that the home ventilation device **130** operates in the home **140**.

(14) Each of the ventilation systems **102** is configured to mechanically move breathable air into and out of lungs in order to assist a patient in breathing. The ventilation systems **102** can provide ventilator data, such as notifications, settings, monitor information (e.g., physiological statistics), and scalars to the hospital ventilation management system **14**. The ventilation system **102** includes a device having appropriate processor, memory, and communications capabilities for processing and providing ventilator data to the hospital ventilation management system **14**. Similarly, the hospital ventilation management system **14** is configured to provide user data, notifications, pre-programmed instructions, lab results, patient data, configuration information, and rules and clinical protocols to each ventilation system **102** in the hospital **101** in order to configure each ventilation system **102** (e.g., remotely over a wired or wireless network, such as LAN **119**). The information provided by the hospital ventilation management system **14** to each ventilation system **102** can be based on the information provided to the hospital ventilation management system **14** by each ventilation system **102**.

(15) For example, a ventilation system **102** can provide the hospital ventilation management system **14** with a current configuration profile and current monitor information for a patient associated with the ventilation system **102**. The hospital ventilation management system **14** can analyze the information provided by the ventilation system **102** in order to determine which modifications, if any, to make to the configuration profile in view of the patient's monitor

information. The hospital ventilation management system **14** may then provide a modified configuration profile to the ventilation system **102** so that the ventilation system **102** may treat the patient in accordance with the modified configuration profile.

(16) The hospital ventilation management system **14** is connected to a wide area ventilation management system **16** configured to manage one or many home ventilation devices **130**. Although the hospital ventilation management system **14** and the wide area ventilation management system **16** are illustrated as being separate systems, both the hospital ventilation management system **14** and the wide area ventilation management system **16** can be hosted or otherwise executed from a single server. In certain aspects, many servers may share the hosting responsibilities of the hospital ventilation management system **14** and the wide area ventilation management system **16**. The server can be any device having an appropriate processor, memory, and communications capability for hosting the hospital ventilation management system **14** and the wide area ventilation management system **16**, and can be in a hospital data center or remotely hosted over a network.

(17) The WAN **120** can include, for example, any one or more of a metropolitan area network (MAN), a wide area network (WAN), a broadband network (BBN), the Internet, and the like. The LAN **119** can include, for example, a personal area network (PAN) or campus area network (CAN). Further, each of the WAN **120** and LAN **119** can include, but is not limited to, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, and the like.

(18) An example use of the ventilator management system will now be provided. A patient associated with the ventilation system **102** is discharged by a clinician **12** from the hospital **101** but still requires ventilation using home ventilation device **130** in the patient's home **140**. The hospital ventilation management system **14** registers with the wide area ventilation management system **16**, and then sends the patient's information and ventilator information from the ventilation system **102** for the patient to the wide area ventilation management system **16**. The home ventilation device **130** is configured using the patient's information and ventilator information and the patient begins treatment using the home ventilation device **130**. The clinician monitors the patient's progress with the home ventilation device **130** by reviewing logs from the home ventilation device **130** that are sent to the hospital ventilation management system **14** through the wide area ventilation management system **16**. As needed, the clinician may modify the configuration parameters of the home ventilation device **130** remotely by sending new configuration parameters from the hospital ventilation management system **14** to the wide area ventilation management system **16**, which then sends the new configuration parameters to the home ventilation device **130** for review by the patient or caregiver. The patient or caregiver accepts the new configuration parameters and the home ventilation device **130** begins to operate according to the new configuration parameters.

(19) FIG. **2** is a block diagram illustrating an example ventilation system **102**, ventilation management system **150**, and home ventilation device **130** from the architecture **10** of FIG. **1** according to certain aspects of the disclosure. Although the ventilation management system **150** is illustrated as connected to a ventilation system **102** and a home ventilation device **130**, the ventilation management system **150** is configured to also connect to infusion pumps, point of care vital signs monitors, and pulmonary diagnostics devices.

(20) The ventilation system **102** is connected to the ventilation management system **150** over the LAN **119** via respective communications modules **110** and **160** of the ventilation system **102** and the ventilation management system **150**. The ventilation management system **150** is connected over WAN **120** to the home ventilation device **130** via respective communications modules **160** and **146** of the ventilation management system **150** and the home ventilation device **130**. The home ventilation device **130** is configured to operate substantially similar to the ventilation system **102** of the hospital **101**, except that the home ventilation device **130** is configured for use in the home **140**. The communications modules **110**, **160**, and **146** are configured to interface with the networks to send and receive information, such as data, requests, responses, and commands to other devices on

the networks. The communications modules **110**, **160**, and **146** can be, for example, modems or Ethernet cards.

(21) The ventilation management system **150** includes a processor **154**, the communications module **160**, and a memory **152** that includes hospital data **156** and a ventilation management application **158**. Although one ventilation system **102** is shown in FIG. **2**, the ventilation management system **150** is configured to connect with and manage many ventilation systems **102**, both ventilation systems **102** for hospitals **101** and home ventilation devices **130** for use in the home **140**.

(22) In certain aspects, the ventilation management system **150** is configured to manage many ventilation systems **102** in the hospital **101** according to certain rules and procedures. For example, when powering on, a ventilation system **102** may send a handshake message to the ventilation management system **150** to establish a connection with the ventilation management system **150**. Similarly, when powering down, the ventilation system **102** may send a power down message to the ventilation management system **150** so that the ventilation management system **150** ceases communication attempts with the ventilation system **102**.

(23) The ventilation management system **150** is configured to support a plurality of simultaneous connections to different ventilation systems **102** and home ventilation devices **130**. The number of simultaneous connections can be configured by an administrator in order to accommodate network communication limitations (e.g., limited bandwidth availability). After the ventilation system **102** successfully handshakes with (e.g., connects to) the ventilation management system **150**, the ventilation management system **150** may initiate communications to the ventilation system **102** when information becomes available, or at established intervals. The established intervals can be configured by a user so as to ensure that the ventilation system **102** does not exceed an established interval for communicating with the ventilation management system **150**.

(24) The ventilation management system **150** can provide the data to the ventilation system **102** in a first-in-first-out (FIFO) order. For instance, if a software upgrade is scheduled to be sent to a ventilation system **102**, the software upgrade can be deployed at configurable timeframes in FIFO order for the specified ventilation systems **102**. Upon receipt, a ventilation system **102** may initialize the software upgrade on a manual reboot. An admit-discharge-transfer communication can be sent to specified ventilation systems **102** within a certain care area of the hospital **101**. A configuration profile communication can be sent to all ventilation systems **102** connected to the ventilation management system **150**. On the other hand, orders specific to a patient are sent to the ventilation system **102** associated with the patient.

(25) The ventilation system **102** may initiate a communication to the ventilation management system **150** if an alarm occurs on the ventilation system **102**. The alarm may be sent to the beginning of the queue for communicating data to the ventilation management system **150**. All other data of the ventilation system **102** may be sent together at once, or a subset of the data can be sent at certain intervals.

(26) The hospital data **156** includes configuration profiles configured to designate operating parameters for the ventilation system **102**, operating parameters of the ventilation system **102** and/or physiological statistics of a patient associated with the ventilation system **102**. Hospital data **156** also includes patient data for patients at the hospital **101**, order (e.g., medication orders, respiratory therapy orders) data for patients at the hospital **101**, and/or user data (e.g., for caregivers associated with the hospital **101**).

(27) The physiological statistics of the ventilator data includes, for example, a statistic for compliance of the lung (Cdyn, Cstat), flow resistance of the patient airways (Raw), inverse ratio ventilation (I/E), spontaneous ventilation rate, exhaled tidal volume (Vte), total lung ventilation per minute (Ve), peak expiratory flow rate (PEFR), peak inspiratory flow rate (PIFR), mean airway pressure, peak airway pressure, an average end-tidal expired CO2 and total ventilation rate. The operating parameters include, for example, a ventilation mode, a set mandatory tidal volume,

positive end respiratory pressure (PEEP), an apnea interval, a bias flow, a breathing circuit compressible volume, a patient airway type (for example endotracheal tube, tracheostomy tube, face mask) and size, a fraction of inspired oxygen (FiO2), a breath cycle threshold, and a breath trigger threshold.
(28) The processor **154** of the ventilation management system **150** is configured to execute instructions, such as instructions physically coded into the processor **154**, instructions received from software (e.g., ventilation management application **158**) in memory **152**, or a combination of both. For example, the processor **154** of the ventilation management system **150** executes instructions to receive ventilator data from the ventilation system **102** (e.g., including an initial configuration profile for the ventilation system **102**).
(29) FIG. **3** illustrates an example flow chart **300** of exchanging data between the ventilation system **102** and the ventilation management system **150**. As illustrated in the flow chart **300**, the ventilation system **102** is configured to send ventilator information, notifications (or "alarms"), scalars, operating parameters **106** (or "settings"), physiological statistics (or "monitors") of a patient associated with the ventilation system **102**, and general information. The notifications include operational conditions of the ventilation system **102** that may require operator review and corrective action. The scalars include parameters that are typically updated periodically (e.g., every 500 ms) and can be represented graphically on a two-dimensional scale. The physiological statistics represent information that the ventilation system **102** is monitoring, and can dynamic based on a specific parameter. The operating parameters **106** represent the operational control values that the caregiver has accepted for the ventilation system **102**. The general information can be information that is unique to the ventilation system **102**, or that may relate to the patient (e.g., a patient identifier). The general information can include an identifier of the version and model of the ventilation system **102**.
(30) In the example of FIG. **3**, the data is sent via a serial connector. The data is sent to a wired adapter **304** having a serial connector and a TCP connector **308**. The data is sent using any appropriate communication protocol **400** (e.g., VOXP protocol). FIG. **4** illustrates an example flow chart for a communication protocol, the VOXP protocol, used by the ventilation system **102** of FIG. **3**.
(31) The communication protocol **400** of FIG. **4** is configured, in certain aspects, to operate in an active mode and a passive mode. In active mode, the ventilation system **102** both responds to requests (e.g., from the ventilation management system **150**), as well as automatically sends data as it becomes available to the ventilation system **102**. In passive mode, the ventilation system **102** responds to requests but does not automatically send data as it becomes available. The protocol **400** begins by transition from a dormant (or "passive") mode **401** to starting the VOXP protocol **402** (e.g., to enter into active mode). When the communication input/output port is ready, a connection is established **403** with the destination (e.g., wired adapter **304**). If the connection is established without a ventilation device **118** being connected to the ventilation system **102**, then the protocol instructs to wait for docking **404** (e.g., of a ventilation device **118**). If a connection is broken while waiting for docking, the link between the ventilation system **102** and the destination is reestablished **405**. Otherwise, when a ventilation device **118** is docked, or a connection is established, the protocol waits for a profile or other data request **406** (e.g., from the ventilation management system **150**). If the connection is broken while waiting for the profile request, the link between the ventilation system **102** and the destination is reestablished. When the profile request is received, ventilation system **102** sends a configuration profile **108** (specifying the capabilities of the ventilation system **102** and the set of operating parameters and other data that it can provide), and then the protocol waits for a configuration command **407** (e.g., from the ventilation management system **150**). When the configuration command is received, a link is established with the destination and the link is configured **408**. If while configuring the link there is a processing error, a mode changes, or the link is restarted, the link is again reestablished **405**. Otherwise, upon

configuring the link **408**, the protocol for the ventilation system **102** may enter a passive mode **409** or active mode **410**. In passive mode **409**, the ventilation management system **150** sends requests, at intervals determined by the ventilation management application **158**, for specified information. At each such request, the ventilation system **102** responds with the specified information **318**, which may include notifications (or "alarms"), scalars, operating parameters **106** (or "settings"), and physiological statistics (or "monitors") of a patient associated with the ventilation system **102**. In active mode **410**, the ventilation system **102** sends specified information **318**, which may include notifications (or "alarms"), scalars, operating parameters **106** (or "settings"), and physiological statistics (or "monitors") of a patient associated with the ventilation system **102**, as each item becomes available. For example an operating parameter **106** is sent when a user of the ventilation system makes a change to a set value. When the ventilation system is turned off, the protocol signals a shutdown **411**. Upon shutting down, the protocol can automatically enter a dormant mode **401** (e.g., after 5 seconds).

(32) Returning to FIG. **3**, the wired adapter **304** is configured to receive **312** the data according to the communication protocol **400** of FIG. **4**, and convert the data from a serial connection format to a TCP connection format. The wired adapter **304** then provides **314** the data in the TCP connection format according to the communication protocol **400** of FIG. **4** to a communication system **302**.

(33) The data is received from the ventilation system **102** through the wired adapter **304** by the communication system **302**. The data may be in a native message format of the ventilation system **102**. The communication system **302** is configured to convert the data into an internal messaging format configured for use with a ventilation management system **150**. The conversion can take place according to the system and method of converting messages being sent between data systems using different communication protocols and message structures described in U.S. patent application Ser. No. 13/421,776, entitled "Scalable Communication System," and filed on Mar. 15, 2012, the disclosure of which is hereby incorporated by reference in its entirety for all purposes. The communication system **302** can include, for example, an interface module for communicating with the wired adapter **304**.

(34) The interface module can include information on the communication protocol **400** (e.g., VOXP protocol) and data structure used by the ventilation system **102** and is configured to both receive messages from and transmit messages between the ventilation system **102** and the ventilation management system **150**. For example, the ventilation management system **150** is configured to provide, through the communication system **302** and the wired adapter **304**, patient data, order data, configuration data, user data, preprogrammed information, vital sign information, rules, notifications, and clinical protocols to the ventilation system **102**. The patient data includes, for example, admit-discharge-transfer data, allergy data, diagnosis data, medication history, procedure history, a patient's name, the patient's medical record number (MRN), lab results, or the patient's visit number. Medication history may include a list of the medications and doses that have been administered to the patient, for example sedative medications, muscle paralytic medications, neural block medications, anti-inflammatory medications. Procedure history may include a list of surgical or other interventional procedures that have been administered, for example cardiothoracic surgery; lung lavage; maxillofacial surgery; chest physiotherapy. The order data includes, for example medication order information, procedure order information for at least one of physical therapy or percussion therapy, sedation order information indicating sedation vacations or modes of ventilator therapy, therapy order information for invasive or non-invasive ventilator therapy, or trial order information for spontaneous breathing trials. The configuration data includes, for example, a patient profile, a user interface configuration, a limit configuration, a notification configuration, or a clinical protocol configuration. The notification configuration can indicate whether certain limits or alerts should be enabled or disabled, and the clinical protocol configuration can be used in a particular area of the hospital **101** (e.g., ICU) and indicate which clinical protocol library should be enabled. A clinical protocol library may include several clinical protocols that may be applicable to

a specified group of patients, for example a spontaneous breathing trial clinical protocol. A clinical protocol may include a set of rules defining actions that the ventilation system **102** should effect in response to events such as a change in patient physiological data, for example a spontaneous breathing trial clinical protocol may include a rule that recommences mandatory ventilation in the event that the patient's rapid shallow breathing index (RSBI) exceeds a set threshold. As another example the spontaneous breathing trial clinical protocol may include a rule that a notification should be provided on display device **114** when the patient has been controlling their own respiration within specified limits for a period of one hour. In certain aspects, the notifications can be generated by the ventilation management system **150** and sent to the ventilation system **102** to alert a caregiver or patient near the ventilation system **102**. The user data includes, for example, an identification of a caregiver or a healthcare institution.

(35) After receiving the ventilator data from the ventilation system **102**, the processor **154** of the ventilation management system **150** is configured to determine, based on the ventilator data, a modification to the initial configuration profile for the ventilation system **102**. In certain aspects, the initial configuration profile is received by the ventilation management system **150** from the ventilation system **102**. The processor **154** of the ventilation management system **150** is further configured to generate a modified configuration profile for the ventilation system **102** based on the determined modification. In certain aspects, the modification to the configuration profile is also determined based on the initial configuration profile of the ventilation system **102**. For example, if the initial configuration profile indicated an average end tidal CO2 level that was considered clinically too low for the patient, the configuration profile could be modified to increase the average end tidal CO2.

(36) In certain aspects, the modification to the configuration profile is also determined based on comparing the physiological statistics of the patient with historical patient data (e.g., stored in the hospital data **156**) to identify a modification to at least one operating parameter of the initial configuration profile, and modify the operating parameter based on the identification. For example, if an apnea interval that, based on historical patient data for many patients at the hospital **101**, was not likely to improve the condition of the patient, then the apnea interval of the configuration profile could be modified by the ventilation management system **150**. As another example, if a specified level of tidal ventilation normalized to patient weight, based on historical patient data for many patients at the hospital **101** with a specified diagnosis, has been associated with a reduced length of hospital stay, then the configuration profile could be modified to adjust pressure support to target this level of tidal ventilation.

(37) The processor **154** of the ventilation management system **150** can be further configured to provide the modified configuration profile to the ventilation system **102** for modifying operating parameters **106** in the memory **104** of the ventilation system **102**. The modified configuration profile **108** is stored in the memory **104** of the ventilation system, and used by the processor **112** of the ventilation system **102** to modify the operating parameters **106** in the memory **104** of the ventilation system. In certain aspects, details regarding the modified configuration profile (e.g., the modifications made to operating parameters, an identification of a clinician responsible for approving the modifications, etc.) are provided for display using the display device **114** of the ventilation system **102**.

(38) The ventilation system **102** includes a processor **112**, the communications module **110**, and a memory **104** that includes operating parameters **106** and a configuration profile **108**. The ventilation system **102** also includes an input device **116**, such as a keyboard, scanner, or mouse, an output device **214**, such as a display, and a ventilation device **118** configured to mechanically move breathable air into and out of lungs in order to assist a patient in breathing according to instructions from the ventilation system **102**. The configuration profile **108** includes one or many configuration profiles for operating the ventilation device **118** of the ventilation system **102**. For example, the configuration profile **108** can include a profile for operating the ventilation device **118** in an

intensive care unit, neonatal intensive care unit, or surgical room, or a profile for operating the ventilation device **118** for patients with a specified respiratory diagnosis, such as ARDS, neuromuscular disease, pneumonia, or post-surgical recovery.

(39) The processor **112** of the ventilation system **102** is configured to execute instructions, such as instructions physically coded into the processor **112**, instructions received from software (e.g., from configuration profile **108**) in memory **104**, or a combination of both. For example, the processor **112** of the ventilation system **102** executes instructions to configure the ventilation device **118**. The processor **112** of the of the ventilation system **102** executes instructions from the configuration profile **108** causing the processor **112** to receive, over the LAN **119**, at least one of patient data, order data, configuration data, or user data. The configuration data can include, for example, an indication (e.g., a set limit) for limiting use of the ventilation system **102** within the hospital **101**. The processor **112** of the of the ventilation system **102** is also configured to provide a modification of operating parameters **106** of the ventilation device **118** based on the received patient data, order data, configuration data, or user data.

(40) In certain aspects the patient data received by the ventilation system **102** includes a patient identifier, such as a MRN, that is obtained through various processes **510**, **520**, and **530** and used to contextualize data generated by the ventilation system **102** as illustrated in FIG. **5**. The contextualization of data includes identifying data generated by a ventilation system **102** as being data associated with a specific patient (a “patient context”). The patient context and ventilation system **102** to patient association can be stored in the memory **103** of the ventilation system **102** or in the hospital data **156** in the memory **152** of the ventilation management system **150**.

(41) As provided in process **510** of FIG. **5**, a ventilation system **102** can be associated with a patient manually when the ventilation system **102** first receives in step **511** an external admit-discharge-transfer alert (e.g., from the ventilation management system **150** or a hospital information system) for a patient. Next, in step **512**, the ventilation system **102** is connected to the patient and in step **513** a caregiver, using input device **116** and display device **114**, searches for the patient's name or identifier (from among a list of patient names/identifiers) on the display device **114** of the ventilation system **102**. The patient's identifier can be found, for example, using a search by care area, patient type, alphabetically, or a list of patients associated with the caregiver. In step **514**, the user validates the patient data (e.g., selects the patient to associate with the ventilation system **102**) and in step **515** the patient is associated with the ventilation system **102**. In certain aspects, a second identifier can be required, such as a medical record number, in order to validate the patient data.

(42) As provided in process **520** of FIG. **5**, a ventilation system **102** can be associated with a patient automatically when the ventilation system **102** again first receives in step **521** an external admit-discharge-transfer alert (e.g., from the ventilation management system **150** or a hospital information system) for a patient and the ventilation system **102** is connected to the patient in step **522**. Next, in step **523**, a clinician performs an electronic search for the patient by, for example, scanning a barcode on the patient's wrist with the input device **116** or having the ventilation system **102** identify the patient using a radio frequency identification (RFID). Next, in step **524**, the user validates the patient data (e.g., confirms the automatically identified patient) and in step **525** the patient is associated with the ventilation system **102**.

(43) As provided in process **530** of FIG. **5**, a ventilation system **102** can also be associated with a patient automatically when the ventilation system **102** is connected to a patient in step **531** and an external system (e.g., a network scanner connected to a server, such as the ventilation management system **150** or an admit-discharge-transfer system) performs a search for the patient (e.g., using RFID). The user in step **533** validates the patient data identified by the external system and the external system sends the patient identification to the ventilation system **102** in step **534**. In step **535** the patient is associated with the ventilation system **102**. As yet another example, a ventilator may first be connected to a patient, the ventilation system **102** or user then performs an electronic

search by, for example, and RFID or scanned patient barcode, the external system validates patient data, the external system sends patient data to the ventilation system **102**, and the patient is associated with the ventilation system **102**.

(44) In certain aspects, both the ventilation management system **150** and ventilation system **102** are configured to cache data, such as the patient data, order data, configuration data, user data, vital sign information (e.g., physiological statistics of a patient), rules, notifications, clinical protocols, and operating parameters. Cached (or "logged") data can be used to perform analytics that result in improved patient care. By caching the data even when the ventilation system **102** or the ventilation management system **150** are not connected, the data will have a greater chance of being used for analytics and result in improved patient care. The data may be cached, for example, when the LAN **119** connection is unavailable. The data may then be shared between the ventilation management system **150** and ventilation system **102** when the connection becomes available. Similarly, the data may then be shared between the ventilation management system **150** and ventilation system **102** at regularly scheduled intervals (e.g., every 30 minutes). The scheduled intervals are configurable by a caregiver or other user, and can be based on, for example, the data being transmitted, when a change is made to an operating parameter of the ventilation system **102**, or when a measured value, alarm threshold, or monitored value reach a predefined level or rate of change. The home ventilation device **130** can also cache data similar to the ventilation system **102**. The data may be cached by the home ventilation device **130**, for example, when the WAN **120** connection is unavailable.

(45) For example, any data that is generated by the ventilation system **102** for documentation, clinical decision support, biomedical engineering or maintenance support can be cached in the memory **104** of the ventilation system **102** to be sent out to the ventilation management system **150**. Similarly, any data that needs to be sent to the ventilation system **102** from the ventilation management system **150** can be cached in memory **152** at the ventilation management system **150** until a scheduled time to send the data, or a next time the ventilation system **150** and ventilation are connected.

(46) FIGS. **6**A and **6**B illustrate example flow charts for caching data on a ventilation system **102** and a ventilation management system **150**. In FIG. **6**A, data **318** for the ventilation system **102**, including ventilation system information, alarms, scalars, settings, and monitors, when available, is sent to the ventilation management system **150** via a connector **316** for storage as hospital data **156** when a connection **602** between the ventilation system **102** and the ventilation management system **150** is available. Otherwise, when the connection **602** between the ventilation system **102** and the ventilation management system **150** is not available, the data is stored in a data cache **604** on the ventilation system **102**.

(47) In FIG. **6**B, data **652** for the ventilation management system **150**, including user data, alerts, preprogrammed information, lab results, patient data, and configuration information, when available, is sent to the ventilation system **102** via a connector **316** for storage as data **658** in memory **104** when a connection **654** between the ventilation system **102** and the ventilation management system **150** is available. Otherwise, when the connection **654** between the ventilation system **102** and the ventilation management system **150** is not available, the data is stored in a data cache **656** on the ventilation management system **150**.

(48) FIG. **7** illustrates an example process **700** for managing a ventilation system using the example ventilation system **102** and ventilation management system **150** of FIG. **2**. While FIG. **7** is described with reference to FIG. **2**, it should be noted that the process steps of FIG. **7** may be performed by other systems.

(49) The process **700** begins by proceeding from beginning step **701** when a ventilation system **102** is initialized and establishes a communication with the ventilation management system **150**, to step **702** when the ventilation system **102** provides ventilator data including at least one of operating parameters of the ventilation device **118** or physiological statistics of a patient associated with the

ventilation device **118** to the ventilation management system **150**. In step **703**, the ventilation management system **150** receives the ventilator data from the ventilation system **102** and in step **704** determines, based on the ventilator data, a modification to the initial configuration profile **108** for the ventilation system **102**. In step **705** a modified configuration profile is generated for the ventilation system **102** based on the determined modification of step **704**, and in step **706** the ventilation management system **706** provides the modified configuration profile to the ventilation system **102** for modifying the operating parameters **106** of the ventilation system **102**. The ventilation management system **706** may also optionally provide at least one of patient data, order data, configuration data, or user data to the ventilation system **102** in step **706**. In step **707**, the ventilation system **102** receives the modified configuration profile and optional patient data, order data, configuration data, or user data. The process **700** then ends in step **708**.

(50) FIG. **7** sets forth an example process **700** for managing a ventilation system using the example ventilation system **102** and ventilation management system **150** of FIG. **2**. An example will now be described using the example process **700** of FIG. **7**.

(51) The process **700** begins by proceeding from beginning step **701** when a ventilation system **102** in the hospital **101** is turned on and establishes a communication with the ventilation management system **150**, to step **702** when the ventilation system **102** provides operating parameters of the ventilation device **118**, physiological statistics of a patient associated with the ventilation device **118**, and an initial configuration profile **108** of the ventilation system **102** to the ventilation management system **150**. In step **703**, the ventilation management system **150** receives the data from the ventilation system **102** and in step **704** determines that the patient's tidal volume has decreased over the last five minutes by 30%, which is an indication of a degradation in the patient's clinical status. The data also indicates the patient's heart rate has increased. The ventilation management system **150** further determines, based on the ventilator data, that the initial configuration profile **108** for the ventilation system **102** should be modified to increase the breath rate parameter. In an alternative example, in step **704** the ventilation management system **150** uses data from other devices such as lab results data **652** including a blood oxygen measurement and a blood carbon dioxide measurement which indicate that the patient is being over-ventilated. The ventilation management system **150** further determines, based on the lab results data, that the initial configuration profile **108** for the ventilation system **102** should be modified to decrease the breath rate parameter. In step **705** the modified configuration profile having the changed breath rate parameter is generated for the ventilation system **102** based on the determined modification of step **704**, and in step **706** the ventilation management system **706** provides the modified configuration profile to the ventilation system **102** for modifying the operating parameters **106** of the ventilation system **102**. The configuration profile **108** of the ventilation system **102** is modified with the modified configuration profile to increase the patient's breath rate, and the process **700** then ends in step **708**.

(52) FIG. **8** is a block diagram illustrating an example computer system **800** with which the ventilation system **102**, ventilation management system **150**, and home ventilation device **130** of FIG. **2** can be implemented. In certain aspects, the computer system **800** may be implemented using hardware or a combination of software and hardware, either in a dedicated server, or integrated into another entity, or distributed across multiple entities.

(53) Computer system **800** (e.g., ventilation system **102**, ventilation management system **150**, and home ventilation device **130**) includes a bus **808** or other communication mechanism for communicating information, and a processor **802** (e.g., processor **112**, **154**, and **136**) coupled with bus **808** for processing information. By way of example, the computer system **800** may be implemented with one or more processors **802**. Processor **802** may be a general-purpose microprocessor, a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated logic, discrete hardware components, or any

other suitable entity that can perform calculations or other manipulations of information.
(54) Computer system **800** can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them stored in an included memory **804** (e.g., memory **104**, **152**, and **132**), such as a Random Access Memory (RAM), a flash memory, a Read Only Memory (ROM), a Programmable Read-Only Memory (PROM), an Erasable PROM (EPROM), registers, a hard disk, a removable disk, a CD-ROM, a DVD, or any other suitable storage device, coupled to bus **808** for storing information and instructions to be executed by processor **802**. The processor **802** and the memory **804** can be supplemented by, or incorporated in, special purpose logic circuitry.
(55) The instructions may be stored in the memory **804** and implemented in one or more computer program products, i.e., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, the computer system **800**, and according to any method well known to those of skill in the art, including, but not limited to, computer languages such as data-oriented languages (e.g., SQL, dBase), system languages (e.g., C, Objective-C, C++, Assembly), architectural languages (e.g., Java, .NET), and application languages (e.g., PHP, Ruby, Perl, Python). Instructions may also be implemented in computer languages such as array languages, aspect-oriented languages, assembly languages, authoring languages, command line interface languages, compiled languages, concurrent languages, curly-bracket languages, dataflow languages, data-structured languages, declarative languages, esoteric languages, extension languages, fourth-generation languages, functional languages, interactive mode languages, interpreted languages, iterative languages, list-based languages, little languages, logic-based languages, machine languages, macro languages, metaprogramming languages, multiparadigm languages, numerical analysis, non-English-based languages, object-oriented class-based languages, object-oriented prototype-based languages, off-side rule languages, procedural languages, reflective languages, rule-based languages, scripting languages, stack-based languages, synchronous languages, syntax handling languages, visual languages, wirth languages, embeddable languages, and xml-based languages. Memory **804** may also be used for storing temporary variable or other intermediate information during execution of instructions to be executed by processor **802**.
(56) A computer program as discussed herein does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, subprograms, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network. The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output.
(57) Computer system **800** further includes a data storage device **806** such as a magnetic disk or optical disk, coupled to bus **808** for storing information and instructions. Computer system **800** may be coupled via input/output module **810** to various devices (e.g., ventilation device **118**). The input/output module **810** can be any input/output module. Example input/output modules **810** include data ports such as USB ports. The input/output module **810** is configured to connect to a communications module **812**. Example communications modules **812** (e.g., communications modules **110**, **160**, and **146**) include networking interface cards, such as Ethernet cards and modems. In certain aspects, the input/output module **810** is configured to connect to a plurality of devices, such as an input device **814** (e.g., input device **116**) and/or an output device **816** (e.g., display device **114**). Example input devices **814** include a keyboard and a pointing device, e.g., a mouse or a trackball, by which a user can provide input to the computer system **800**. Other kinds of input devices **814** can be used to provide for interaction with a user as well, such as a tactile input

device, visual input device, audio input device, or brain-computer interface device. For example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, tactile, or brain wave input. Example output devices **816** include display devices, such as a LED (light emitting diode), CRT (cathode ray tube), or LCD (liquid crystal display) screen, for displaying information to the user.

(58) According to one aspect of the present disclosure, the ventilation system **102**, ventilation management system **150**, and home ventilation device **130** can be implemented using a computer system **800** in response to processor **802** executing one or more sequences of one or more instructions contained in memory **804**. Such instructions may be read into memory **804** from another machine-readable medium, such as data storage device **806**. Execution of the sequences of instructions contained in main memory **804** causes processor **802** to perform the process steps described herein. One or more processors in a multi-processing arrangement may also be employed to execute the sequences of instructions contained in memory **804**. In alternative aspects, hard-wired circuitry may be used in place of or in combination with software instructions to implement various aspects of the present disclosure. Thus, aspects of the present disclosure are not limited to any specific combination of hardware circuitry and software.

(59) Various aspects of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. The communication network (e.g., local area network **119** and wide area network **120**) can include, for example, any one or more of a personal area network (PAN), a local area network (LAN), a campus area network (CAN), a metropolitan area network (MAN), a wide area network (WAN), a broadband network (BBN), the Internet, and the like. Further, the communication network can include, but is not limited to, for example, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, or the like. The communications modules can be, for example, modems or Ethernet cards.

(60) Computing system **800** can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. Computer system **800** can be, for example, and without limitation, a desktop computer, laptop computer, or tablet computer. Computer system **800** can also be embedded in another device, for example, and without limitation, a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, a video game console, and/or a television set top box.

(61) The term “machine-readable storage medium” or “computer readable medium” as used herein refers to any medium or media that participates in providing instructions or data to processor **802** for execution. Such a medium may take many forms, including, but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical disks, magnetic disks, or flash memory, such as data storage device **806**. Volatile media include dynamic memory, such as memory **804**. Transmission media include coaxial cables, copper wire, and fiber optics, including the wires that comprise bus **808**. Common forms of machine-readable media include, for example, floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH EPROM,

any other memory chip or cartridge, or any other medium from which a computer can read. The machine-readable storage medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them.

(62) As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

(63) Furthermore, to the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

(64) A reference to an element in the singular is not intended to mean “one and only one” unless specifically stated, but rather “one or more.” All structural and functional equivalents to the elements of the various configurations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

(65) While this specification contains many specifics, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of particular implementations of the subject matter. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(66) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the aspects described above should not be understood as requiring such separation in all aspects, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(67) The subject matter of this specification has been described in terms of particular aspects, but other aspects can be implemented and are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous. Other variations are within the scope of the following claims.

(68) These and other implementations are within the scope of the following claims.

Claims

1. A ventilator management system, comprising: a memory storing ventilator configuration data, patient physiological statistics, and notifications; and one or more processors configured to: communicatively connect a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available; receive a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator; cache, in the memory, the command while the ventilator is unavailable and provide the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device; receive, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable; provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the patient at periodic intervals; receive the first ventilation data from the ventilator responsive to the requests for first ventilation data; and provide, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.
2. The ventilator management system of claim 1, wherein the one or more processors are further configured to: receive an indication that a ventilator is available for communication in the passive mode; wherein the requests for first ventilation data are provided based on the indication; switch the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and receive, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.
3. The ventilator management system of claim 2, wherein in the passive mode, the one or more processors determine data communication intervals for the ventilator by sending the requests for first ventilation data, and wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.
4. The ventilator management system of claim 1, wherein the one or more processors are further configured to: receive, from the remote computing device, a protocol configuration for a particular care area of a hospital; select, for the ventilator based on the received protocol configuration, from a plurality of protocol libraries, a protocol library applicable to a group of patients associated with the particular care area; and cause the ventilator to be configured with a set of rules associated with the selected protocol library, the set of rules defining actions that the ventilator performs in response to a predetermined change in patient physiological data.
5. The ventilator management system of claim 1, wherein the one or more processors are further configured to: compare the physiological statistic of the patient with historical patient data; identify a modification the configuration parameter of the ventilator based on the comparing; and cause the ventilator to modify the configuration parameter according to the modification.
6. The ventilator management system of claim 1, wherein the one or more processors are further configured to: receive an indication to set specified limits on a respiration of the ventilator; and provide a notification for display when the patient has been controlling the patient's own respiration within specified limits for a predetermined period of time.
7. The ventilator management system of claim 1, wherein the one or more processors are further configured to: provide, to the ventilator, before the configuration parameter is modified, an indication that the patient accepts the configuration parameter.

8. The ventilator management system of claim 1, wherein the one or more processors are further configured to, before the physiological statistic of a patient is received: receive a patient identifier associated with the patient; prompt, after receiving the patient identifier, a clinician for a confirmation to associate the patient with the ventilator; receive the confirmation responsive to prompting the clinician; and associate the ventilator with the patient based on the confirmation, wherein the physiological statistic of a patient is received based on the association of the patient with the ventilator.

9. The ventilator management system of claim 8, wherein the one or more processors are further configured to, before the physiological statistic of a patient is received: receive a search for list of patients associated with a care area or a clinician; provide the list of patients for display; and receive, from the list of patients, a selection of the patient associated with the patient identifier.

10. The ventilator management system of claim 1, wherein the one or more processors are further configured to: receive a physiological parameter associated with the patient from the remote computing device; determine, based on the physiological parameter, that the patient is not receiving a correct amount of ventilation from the ventilator; and cause, responsive to determining that the patient is not receiving the correct amount of ventilation, the ventilator to adjust an operating parameter associated with a breath rate, wherein the modified configuration profile is configured to cause the ventilator to adjust an amount of ventilation provided to the patient based on the adjusted operating parameter.

11. The ventilator management system of claim 1, wherein the remote computing device is a mobile device.

12. A machine-implemented method, comprising: communicatively connecting a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available; receiving a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator; caching, in a memory, the command while the ventilator is unavailable and providing the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device; receiving, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable; and provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the patient at periodic intervals; receive the first ventilation data from the ventilator responsive to the requests for first ventilation data; providing, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.

13. The machine-implemented method of claim 12, further comprising: receiving an indication that a ventilator is available for communication in the passive mode; wherein the requests for first ventilation data are provided based on the indication; switching the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and receiving, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.

14. The machine-implemented method of claim 13, comprising: determining data communication intervals for the ventilator by one or more processors external to the ventilator by sending the requests for the first ventilation data, and wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.

15. The machine-implemented method of claim 12, further comprising: receiving, from the remote computing device, a protocol configuration for a particular care area of a hospital; selecting, for the ventilator based on the received protocol configuration, from a plurality of protocol libraries, a protocol library applicable to a group of patients associated with the particular care area; and causing the ventilator to be configured with a set of rules associated with the selected protocol library, the set of rules defining actions that the ventilator performs in response to a predetermined change in patient physiological data.
16. The machine-implemented method of claim 12, further comprising: comparing the physiological statistic of the patient with historical patient data; identifying a modification the configuration parameter of the ventilator based on the comparing; and causing the ventilator to modify the configuration parameter according to the modification.
17. The machine-implemented method of claim 12, further comprising: receiving an indication to set specified limits on a respiration of the ventilator; and providing a notification for display when the patient has been controlling the patient's own respiration within specified limits for a predetermined period of time.
18. The machine-implemented method of claim 12, further comprising: providing, to the ventilator, before the configuration parameter is modified, an indication that the patient accepts the configuration parameter.
19. The machine-implemented method of claim 12, further comprising, before the physiological statistic of a patient is received: receiving a patient identifier associated with the patient; prompting, after receiving the patient identifier, a clinician for a confirmation to associate the patient with the ventilator; receiving the confirmation responsive to prompting the clinician; and associating the ventilator with the patient based on the confirmation, wherein the physiological statistic of a patient is received based on the association of the patient with the ventilator.
20. The machine-implemented method of claim 19, further comprising, before the physiological statistic of a patient is received: receiving a search for list of patients associated with a care area or a clinician; providing the list of patients for display; and receiving, from the list of patients, a selection of the patient associated with the patient identifier.
21. The machine-implemented method of claim 12, further comprising: receiving a physiological parameter associated with the patient from the remote computing device; determining, based on the physiological parameter, that the patient is not receiving a correct amount of ventilation from the ventilator; and causing, responsive to determining that the patient is not receiving the correct amount of ventilation, the ventilator to adjust an operating parameter associated with a breath rate, wherein the modified configuration profile is configured to cause the ventilator to adjust an amount of ventilation provided to the patient based on the adjusted operating parameter.
22. A non-transitory machine-readable storage medium comprising machine-readable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: communicatively connecting a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available; receiving a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator; caching, in a memory, the command while the ventilator is unavailable and providing the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device; receiving, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable; and provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the

patient at periodic intervals; receive the first ventilation data from the ventilator responsive to the requests for ventilation data; providing, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.

23. The non-transitory machine-readable storage medium of claim 22 comprising machine-readable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving an indication that a ventilator is available for communication in the passive mode, wherein the requests for first ventilation data are provided based on the indication; switching the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and receiving, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.

24. The non-transitory machine-readable storage medium of claim 23 comprising machine-readable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: wherein in the passive mode, the one or more processors determine data communication intervals for the ventilator by sending the requests for first ventilation data, and wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.